

PROJECT AND TEAM INFORMATION

Project Title

Multi-Lane Parking Allocation and Rerouting Simulator

Student / Team Information

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PROPOSAL DESCRIPTION

Motivation

Urbanization has increased at a staggering rate in India in the last few decades. This dramatic increase in urban population is caused by the increase in economic opportunities, better infrastructure, and access to essential services in cities. However, due to this rapid increase in urbanization, there is an increase in privately owned vehicles. As populations grow and incomes rise, more individuals are opting for personal transportation, leading to a surge in the number of vehicles on city roads.

This growth in the number of vehicles has outpaced the development of supporting infrastructure, especially when it comes to parking spaces. As a result, the majority of congestion on roads is not caused by transit, and the real culprit is vehicles looking for parking. Some studies show that almost 30% of urban traffic in congested areas can be attributed to vehicles looking for parking.

This unavailability of parking has a high impact on the environment. As the fuel used by these "parking cruisers" (which is 30-40 % of vehicles on the road) racks up a huge amount of carbon emissions. A study shows that in the United States alone, vehicles searching for parking waste approximately 930 million gallons of gasoline annually. This is the same amount of CO₂ emissions as electricity use in 1.25 million homes.

State of the Art / Current solution

Today, most modern parking systems try to solve this issue by installing sensors over parking spots and placing digital display boards at the entrance to show how many spots are left on each floor. Some facilities also offer mobile apps where drivers can check availability or book a spot in advance.

However, these existing systems are quite basic in how they actually manage vehicles once they enter. They usually assign spots on a simple "first come, first served" basis or just pick whatever spot is physically the closest. Because they do not track how crowded an internal driving lane is getting, multiple cars often end up being sent down the exact same narrow lane at the same time, creating bumper-to-bumper gridlock inside the facility. Furthermore, their navigation systems treat every path as completely clear and cannot dynamically reroute cars when a lane gets blocked or slowed down. Online reservations are also handled in rigid, fixed time blocks rather than exact time windows, which leaves spaces sitting empty between bookings. In short, while current systems are good at showing whether a spot is empty, they fail to coordinate the active movement and flow of cars inside the facility.

Project Goals and Milestones

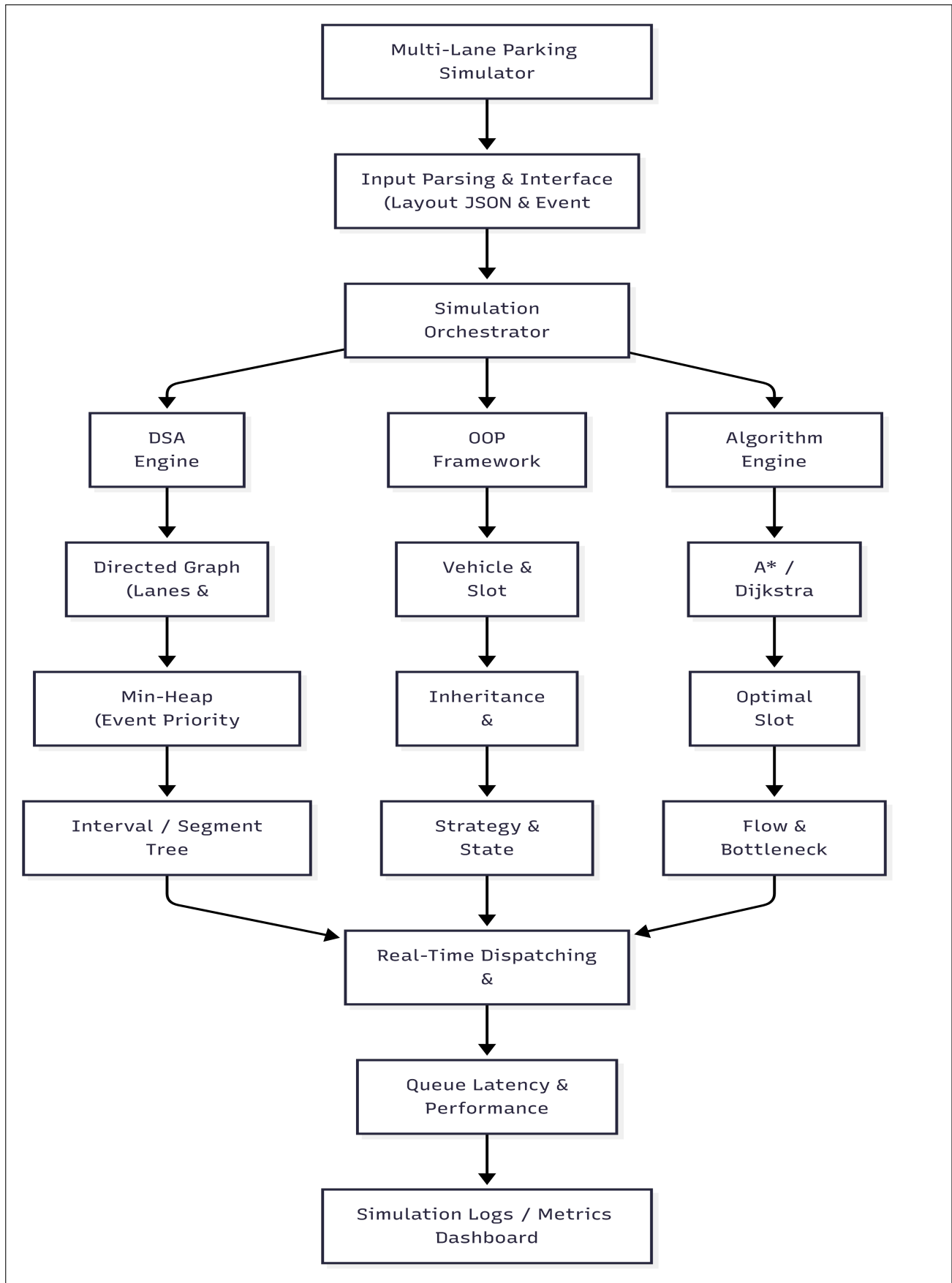
The initial goal of this project is to make a more efficient parking space and simulate how vehicles would navigate through a filled multi-lane parking space to find the nearest spot without wasting any time by roaming to find an open spot that has appropriate space for their vehicle. The project would help reduce the time taken by people to navigate to an empty parking spot, and also reduce the carbon emissions caused by reducing the fuel consumed in travelling to an empty parking spot.

The project would be able to show how different layouts of the parking model would work for different random vehicles, and would generate a parking ticket or fee for parking based on the type of vehicle that is parked(based on space required by the vehicle/EV charging support required, etc) and display it and store it for the admin of the parking lot.

Project Approach

To solve this problem, we will implement an event-driven simulation in C++. We will simulate a multi-floor parking facility. The simulation will start by reading a layout file of the parking space that maps out the entire parking facility, including entry gates and different types of spots. Events such as vehicles arriving and leaving, and temporary closures of lanes are processed in the exact order that they occur using a priority queue. When a vehicle arrives, the system checks its details, such as vehicle size, EV status, or accessibility needs, to find a suitable space. To make navigation easier, specific areas are dedicated to particular needs, placing accessible spots near main exits or elevators and guiding electric cars directly to the spots with charging stations if available. To guide drivers along the best path, we will use an A path-finding algorithm with dynamic weights that would also make the vehicles avoid crowded aisles. For reservations, the system uses interval trees so that the drivers can reserve exact time windows down to the minute without any of the slots overlapping. The engine also keeps account of the total parking duration to track operational billing and calculate revenue generated over time. To keep the code base clean and easy to maintain, we rely on standard object-oriented patterns.*

System Architecture (High Level Diagram)



Project Outcome / Deliverables

- *An engine that would simulate the flow of traffic, arrival and departure of vehicles, parking spots, and lane closures (if any) in real time within the layout of the facility.*
- *A navigation system that guides vehicles through the fastest possible path to the exit or their assigned parking spot, while accounting for blocked or crowded lanes.*
- *A booking tool that would allow the drivers to reserve parking spaces for time windows of their choice without conflicts in the schedule.*
- *A system that checks the requirements of the vehicles, assigns them to dedicated zones and calculates the bill and revenue based on the parking duration.*
- *A module that generates detailed reports on various administrative things, such as the overall facility revenue, spot occupancy rate, etc.*
- *A configuration loader that allows the admin to test custom map layouts, supported by test suites and software documentation.*

Assumptions

We have made the following assumptions:

- *Drivers follow the instructions that are provided to them, parking only in the spots assigned to them and following the route provided.*
- *The vehicles arrive and depart within the timestamps provided during the booking.*
- *The movement inside the parking area is free of any erratic driving.*
- *The provided layout files are in the correct format and have valid time stamps and coordinates.*
- *Every vehicle provides all the necessary information on arrival, such as vehicle ID, license plate, etc.*

References

[1] Shubham, A. Kumar, and S. Kumar, "AI-Powered Smart Parking: Reducing Traffic Congestion Through Intelligent Allocation," Int. J. Res. Trends Innov., vol. 10, no. 4, pp. b557–b569, Apr. 2025.

[2] <https://parknav.com/hidden-climate-cost-of-finding-parking-space>